Measuring the Price of Causality in Low-Latency Polyphonic Guitar Transcription

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***Abstract*—State-of-the-art automatic music transcription systems for guitar achieve high accuracy but rely on non-causal architectures. This letter presents a causal, streaming guitar-to-MIDI system operating on direct-input (DI) signals with an 8.8 ms expected algorithmic delay (11.5 ms worst-case). The system processes each 5.3 ms hop in under 4.3 ms on a single CPU core. Evaluated zero-shot on the EGDB electric-DI corpus, the proposed causal model achieves a note onset F-measure of 0.744 ($\pm$50 ms tolerance). This performance exceeds the standard offline baseline Basic Pitch (0.722) and is 0.084 below a 20.2-million-parameter offline convolutional recurrent neural network. To isolate the performance cost of causality, we introduce a non-causal twin architecture initialized with causal weights. The twin closes the performance gap (0.828 vs. 0.829), demonstrating that the causality constraint costs at most approximately 0.084 in onset F-measure. An analysis of the EGDB dataset reveals that at least 22 % of all onset labels (28 % of matched notes) are annotated more than 10 ms after the acoustic attack, which bounds strict-tolerance scores for every system.**

***Index Terms*—Automatic music transcription, causal systems, electric guitar, low latency, real-time audio.**

I. Introduction

Playing a virtual synthesizer live through a guitar requires an automatic music transcription (AMT) system that operates with minimal latency and strict causality. The system must process audio without future context and complete signal analysis, inference, and decoding within a few milliseconds on a host CPU. Recent advancements in guitar AMT largely focus on offline applications, leveraging extensive non-causal context to maximize accuracy [1], [2], [3]. While some real-time AMT systems exist, they either report high latencies, such as 360 ms [4], or lack comprehensive benchmarks and latency specifications. Open-source real-time systems like GuitarMidi-LV2 report pitch-dependent latencies (4–16 ms) that are highest at lower frequencies [5]. Currently, no study provides a detailed algorithmic latency budget alongside benchmark accuracy on a public guitar corpus.

The causality constraint significantly impacts performance. For piano AMT, imposing strict causality and low latency reduced the onset F-measure of a state-of-the-art system from 0.668 to 0.374 [6]. In this letter, we quantify the trade-off between causality and transcription accuracy for electric guitar DI signals. The main contributions are:

- A causal, streaming DI-to-MIDI transcription system (2.18 M parameters) with a detailed algorithmic delay breakdown (8.8 ms expected, 11.5 ms worst-case) and per-frame compute measured on a single CPU core.
- A zero-shot benchmark of a causal guitar transcription system on the public EGDB corpus [1], compared against offline baselines using identical audio and evaluation protocols.
- An empirical measurement of the performance cost of causality using a non-causal twin of the proposed architecture.
- An analysis of annotation timing in the EGDB dataset, demonstrating that delayed onset labels fundamentally limit strict-tolerance evaluation scores.

The training pipeline, a real-time Windows inference application, and a live demonstration video are publicly available.[1]

II. System

A. Causal Front-End

The input is the guitar DI signal sampled at 24 kHz. Two causal Short-Time Fourier Transforms (STFTs) with window lengths of 2048 and 512 samples share a 128-sample hop size. The spectra are projected onto a constant-Q grid of 256 log-frequency bins (36 bins per octave, C2 to 9 kHz). This representation ensures that pitch shifts correspond to translations along the frequency axis, allowing a learned harmonic template to transfer across registers. Each frame is factorized into a gain-invariant spectral shape and a relative level based on a $\sim$1 s running estimate, making the features robust to static input gain variations.

Algorithmic delay originates from two primary sources (Fig. 1). First, *framing* incurs an expected delay of 2.67 ms (5.33 ms worst-case) while waiting for the current hop to fill; the exact delay is determined by where within the hop the attack lands, so these figures are analytic properties of the hop size, not empirical means. Second, *detection* delay arises because a causal window applies the least weight to its newest samples; an acoustic attack must propagate into the heavily weighted region to be detected. To mitigate this, the 512-sample transient window uses an asymmetric shape peaking at 85 % of its length. This achieves 50 % broadband visibility 6.12 ms after an attack, compared to $\sim$10.7 ms for a symmetric Hann window. The total algorithmic delay is 8.79 ms expected (11.45 ms worst-case) for broadband attacks. The 2048-sample window provides low-frequency pitch resolution and does not influence onset timing. These values represent algorithmic delay and are independent of hardware buffers or driver configurations.

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[1] https://github.com/maximmashkov1/gdi2m

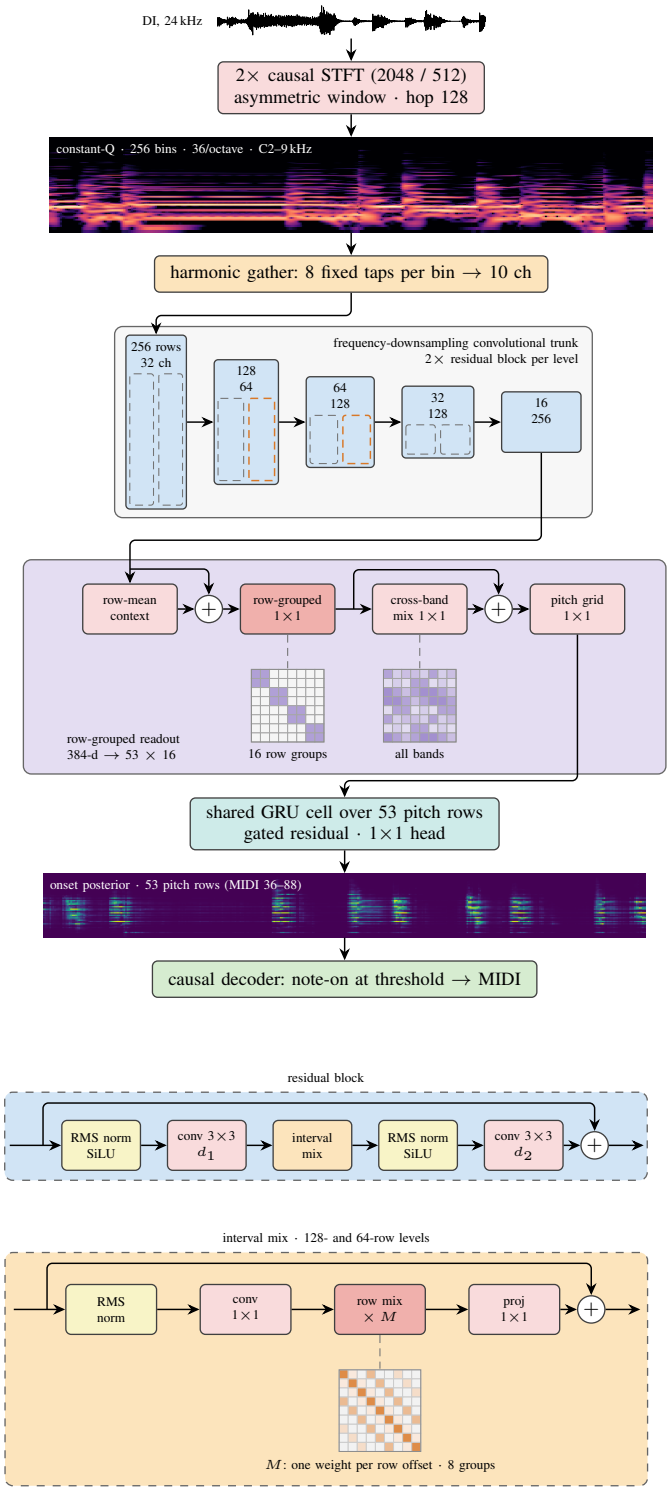


Fig. 1. Signal path. All image panels are real model I/O for a 2 s excerpt of EGDB DI item 50 through the evaluated checkpoint. Algorithmic delay is 2.67 ms expected framing + 6.12 ms transient visibility = 8.79 ms expected (11.45 ms worst-case; 12.50 ms on tonal content); the decoder adds zero delay. Single-thread CPU compute is 3.15 ms mean (4.24 ms max) per 5.33 ms hop. Residual-block time-axis convolutions are causal (left-padded), frequency-axis centered; dilations (d_1, d_2) are $(1, 1)$ in a level's first block, $(2, 4)$ in its second. Orange slots mark the two residual blocks that carry the interval mix.

B. Network Architecture and Decoder

The acoustic model features a four-level 2D convolutional trunk that halves the frequency resolution at each level without downsampling in time, providing a causal receptive field of 357 ms. Before mixing, each frequency bin row aggregates its corresponding overtone series using a fixed 8-tap harmonic gather operation. Because the constant-Q representation fixes harmonic intervals to constant row offsets, cross-frequency mixing is performed via interval-indexed weights (applied as Toeplitz matrices), followed by a global row-mean context layer. This fixed interval geometry replaced softmax attention over frequency during development. After a row-grouped bottleneck readout, a shared Gated Recurrent Unit (GRU) cell processes each of the 53 pitch rows independently. This cell utilizes a learned, L1-regularized residual gate to maintain memory of sustained notes that are no longer prominent in the spectrum. A shared linear head outputs onset and sustain probabilities per 128-sample hop.

The decoding process is causal and operates on a per-frame basis. A note-on event is triggered in the frame in which its onset probability exceeds a predefined threshold (0.85, determined via internal validation), utilizing Schmitt-trigger re-arming and a 10.7 ms refractory period per pitch row. Note-off events follow the sustain head output with a 42.7 ms debounce period. No future confirmation window is used (Sec. V-C).

C. Training Data

The model is trained on three public corpora: GOAT [7] and Guitar-TECHS [8] (electric DI), and GuitarSet [9] (acoustic, training only), totaling approximately 10 hours of audio across 545 recordings and 10 players. Training windows are composed rather than sampled: a dataset-weighted base window is pitch-shifted by resampling within per-recording limits (hexaphonic sources also receive per-string gain remixes); a second window may be overlaid, and isolated note or chord units from an event-sliced sample bank pasted at random offsets. Labels merge by union, kept exact by the pitch-major representation. Composed windows pass through a randomized DI timbre chain (pickup-position comb, pickup-resonance biquad, smooth random equalization, bounded soft clipping), then level jitter, mains hum and hiss, and a measured idle noise floor. Two rules bound the strategy: every operation that alters note content returns an exact label transform, and every window must remain a plausible direct-input signal, which excludes amplifier and cabinet rendering. The corpora's onset annotations sit systematically late relative to the physical acoustic attack; these lags are estimated per corpus and removed at ingest, training the model to detect the attack itself. The EGDB dataset [1] is strictly reserved as an external test set and never enters the training pipeline.

III. Evaluation Protocol

Corpus and Metrics. We evaluate performance using the EGDB dataset [1], comprising 240 electric guitar DI recordings (107.5 minutes, 35,700 notes). The system is evaluated on the DI audio against the original, unmodified labels using the standard `mir_eval` library [10]. The primary metric is the micro-averaged note onset F-measure at a ±50 ms tolerance, with a 50-cent pitch tolerance. Following the EGDB authors'

methodology, we evaluate onsets only, as offset labels are considered unreliable [1]. Additional evaluations report F-measures at 10, 20, and 30 ms tolerances (Fig. 2). Audio is normalized per clip to a standard input level (−18 dBFS), an operation that utilizes no label information and is applied uniformly across evaluated systems; it shifts F1 by −0.003 for the proposed system and −0.001 for Basic Pitch, so no comparison depends on it. Two data subsets are reported: all 240 clips for zero-shot evaluation, and clips 216–240 corresponding to the original EGDB test split [1].

Baselines. Basic Pitch 0.4.0 [11], a standard instrument-agnostic AMT system, is used as an open-source offline baseline and evaluated on identical audio. For published offline baselines, we compare against models trained on EGDB clips 1–215 and tested in-domain on clips 216–240 [1], as well as a 20.22 M-parameter convolutional recurrent neural network (CRNN) [12] evaluated zero-shot on all 240 clips [3]. This CRNN incorporates bidirectional GRUs over 10 s segments and is fine-tuned on GAPS [13] and GOAT corpora; GOAT also appears in our training mix, making this the closest matched offline reference available. We exclude baselines trained exclusively on piano or orchestral data, as evaluating them on guitar DI primarily measures domain mismatch rather than transcription efficacy.

IV. The Non-Causal Twin

The performance gap between a real-time system and an offline system conflates the impact of causality with differences in model capacity, architecture, and training data. To isolate the effect of causality, we developed a non-causal twin of the proposed architecture. The modification follows a strict single-variable principle: any architectural improvement applicable to both models is excluded from the twin to strictly measure the utility of future context.

Time-axis convolutions are expanded from causal kernels ($k = 3$) to centered kernels ($k = 5$). At dilation d, the causal kernel processes time steps $\{t-2d, t-d, t\}$, whereas the centered kernel processes $\{t-2d, \ldots, t+2d\}$, preserving the identical backward receptive field while adding an equivalent forward span. Additionally, the GRU incorporates a backward branch with a zero-initialized gate, leaving the forward path unchanged. Because the causal taps map perfectly to the first three taps of the centered kernels, the twin is initialized exactly to the causal model's weights. The twin is then fine-tuned for 30,000 steps from this identical starting point, on the unchanged data and augmentation pipeline, which removes initialization variance from the comparison; the additional training budget itself remains a confound, which is why the measured gain is an upper bound. The resulting twin model contains 3.09 M parameters (+42 %) and a centered convolutional window of ±66 frames (±352 ms), mirroring the causal 67-frame window. This non-causal model is designed strictly for offline evaluation.

V. Results

A. Overall Performance

As shown in Table I, the proposed causal system achieves an onset F-measure of 0.7444 across all 240 EGDB clips.

TABLE I
Note Onset F1 (±50 ms, micro) on EGDB DI

system	params	causal	context	F1
zero-shot, all 240 clips				
ours, causal	2.18 M	yes, 8.8 ms	357 ms past	0.7444
ours, non-causal twin	3.09 M	no	±352 ms	0.8281
Basic Pitch 0.4.0	0.017 M	no	2 s window	0.7221
Kong CRNN [3]	20.22 M	no	10 s biGRU	0.8286
+ velocity pretrain [3]	20.22 M	no	10 s biGRU	0.8295
clips 216–240 (2022 test split)				
ours, causal (same run)	2.18 M	yes	357 ms past	0.7814
Basic Pitch 0.4.0 (same run)	0.017 M	no	2 s window	0.7793
multi-loss Transformer† [1]	6.01 M	no	∼3 s	0.613
Onsets & Frames† [14], [1]	0.74 M	no	—	0.591

Raw labels, stock `mir_eval`. Upper block: zero-shot, all 240 clips. Lower block: the 2022 test split, a different experiment. Quoted rows keep their sources' protocols. †Trained on EGDB clips 1–215, tested in-domain.

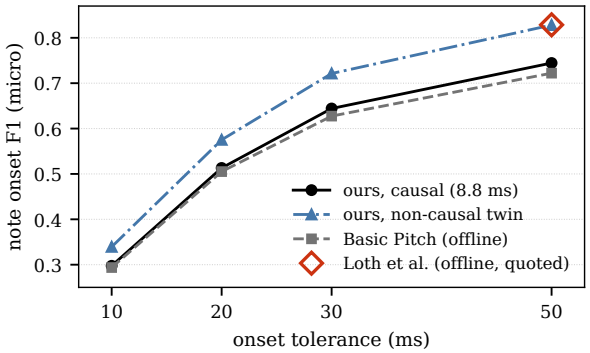


Fig. 2. Onset F1 against matching tolerance, all 240 EGDB clips. At 10 ms the causal system and the offline baseline perform similarly (0.2975 vs. 0.2937); the twin shows the column is not purely an annotation ceiling (0.3393). Section V-B bounds the labels' own timing.

This performance surpasses the offline Basic Pitch baseline (0.7221) evaluated on identical audio, and is 0.084 below the larger 20.22 M-parameter offline CRNN. On the EGDB 2022 test split (clips 216–240), the causal system achieves 0.7814, outperforming the in-domain 2022 baselines (0.591–0.613); this margin is attributable primarily to the larger training corpora now available rather than to architecture, as the 2022 models were trained solely on 215 EGDB clips from a single guitarist. Analysis of failure modes indicates that out-of-domain, the causal model tends to over-predict (precision 0.695, recall 0.802). Basic Pitch exhibits a similar trend (precision 0.655, recall 0.804). Excluding 65 clips with identified annotation defects (e.g., exact quantized grids or wholesale temporal shifts) alters the overall F-measure by only 0.0025, confirming the robustness of the evaluation against corpus errors.

B. Analysis of Annotation Timing

Evaluating systems at strict temporal tolerances (e.g., 10 ms) requires reliable ground-truth timing. Analysis reveals that 46.3 % of the causal system's true positive matches (at ±50 ms tolerance) occur *before* the corresponding reference label. The signed delay distribution shows a mean of +1.3 ms and a 10th percentile of −24.7 ms. This early fraction is not a matching artifact: restricted to reference notes with no neighbor within ±300 ms, 44.2 % of matches remain early and 25.2 % are early by more than 10 ms (28.3 % in dense passages), and

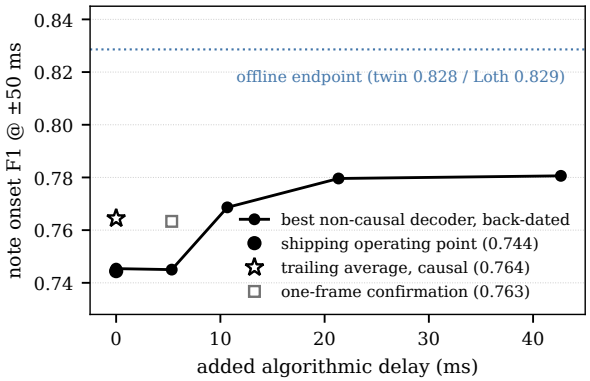


Fig. 3. Onset F1 at ±50 ms against added algorithmic delay for decoders operating on the same causal model's output. The offline endpoint marks the shared level of the non-causal twin and the 20.22 M offline CRNN [3].

the offline Basic Pitch exhibits the same distribution shape around the same labels (51.8 % early matches, 10th percentile −25.9 ms). Since a causal system cannot physically anticipate an acoustic attack, and the front-end reaches half-visibility only at +6.1 ms, a match more than 10 ms early implies a label more than 10 ms late: 28.0 % of matched notes, hence at least 22.5 % of all EGDB onset labels, are annotated more than 10 ms after the actual acoustic onset. This late-annotation bias limits the maximum achievable score for any system at strict tolerances (Fig. 2). Scores are reported against the raw labels, unshifted, so every system is compared on the same reference.

C. Ablation of Causal Constraints

Figure 3 decomposes the performance gap to offline systems by modifying the decoder's lookahead on the causal model's output. Traditional peak-picking reduces performance across all lookahead values (1 to 32 frames). Probabilistic smoothing with live-realizable timestamps improves the F-measure by up to +0.027. However, a purely causal trailing average recovers +0.020 of this gain, indicating that the benefit primarily stems from denoising rather than future context. Even with back-dated note-on times, as plotted, the decoder's non-causal upper bound is +0.036. One frame of confirmation would gain +0.019 for 5.33 ms of delay, so it is not used.

The remaining gap is attributable to the acoustic model itself. The non-causal twin achieves an F-measure of 0.8281 using the same causal decoder. This result closely matches the offline CRNN's score of 0.8286, demonstrating that the offline performance endpoint is replicable despite significant differences in model capacity (3.09 M vs. 20.22 M parameters) and context length (±0.35 s vs. 10 s segments). Compared against this replicated offline endpoint, the causality constraint incurs a maximum cost of approximately 0.084 in onset F-measure.

VI. Limitations and Conclusion

Limitations of the proposed system include a tendency to over-predict notes on out-of-domain data (a precision bound) and an inability to distinguish unison notes played simultaneously across multiple strings, as they map to a single pitch row. Additionally, evaluation is constrained by annotation delays inherent in the test corpus, which limits the reliability of strict-tolerance metrics. External validation is further limited to a single corpus, whose DI signal is a summed hexaphonic pickup rather than a conventional magnetic pickup; offsets are not evaluated, following the corpus authors' own practice.

In conclusion, our causal, streaming guitar-to-MIDI transcription system achieves a 0.744 onset F-measure on the public EGDB corpus. Operating with an 8.8 ms algorithmic delay and consuming less than 4.3 ms of compute per hop on a single CPU core, the system meets strict real-time processing constraints. By comparing against a controlled non-causal twin architecture, we demonstrate that the causality constraint imposes a quantifiable performance penalty of $\leq$0.084 in F-measure relative to strong offline systems evaluated under the identical protocol.

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